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Prof. Kenneth M. Merz, Jr., Editor-in-Chief
Journal of Chemical Information and Modeling
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Dear Prof. Merz,

We submit our manuscript **"Resistance-Aware Computational Analysis of Antimalarial Leads from African Natural Products"** for consideration as an Article in *JCIM*.

This work presents a resistance-aware computational analysis that combines docking-based RRS, ACSI, and PNS profiling of 17 set-C polypharmacology candidates with a separate targeted MD check of four named parent-study consensus complexes (10 ns each; 40 ns total). The 17 set-C candidates were evaluated against six clinically relevant mutations (PfDHFR-N51I, C59R, S108N, I164L; PfCRT-K76T, K76A) by docking; they were not subjected to the parent-study production MD. Only PfCRT–214 yielded an interpretable MM-GBSA endpoint estimate, while the other parent-study systems were dissociated or conversion-excluded. External Tartarus analysis reports orthogonality between MPO and raw docking affinity (Spearman $\rho \approx 0$, $p = 0.09$), not independence among MPO components.

The code, compact scoring outputs, and source-data manifests are available in the GitHub repository under an open-source licence. Large parent-study trajectories are identified with their analysis boundaries and are not used beyond the systems and checksums that can be verified. Statistical robustness is addressed through confidence intervals and Bonferroni-corrected hypothesis testing.

Suggested reviewers.

1. Dr. Fidele Ntie-Kang, University of Buea — African NP computational discovery
2. Dr. Kelly Chibale, University of Cape Town — antimalarial drug discovery
3. Dr. John D. Chodera, Memorial Sloan Kettering — free energy methods, MD best practices

This manuscript is original, not under consideration elsewhere, and all authors have approved the submission.

Sincerely,

Myke-Vital Temgoua Sao
(on behalf of all co-authors)